

Introduction to Computer Organisation and Architecture UCCD1133



Chapter 6

Processor, Memory System and Instruction Execution

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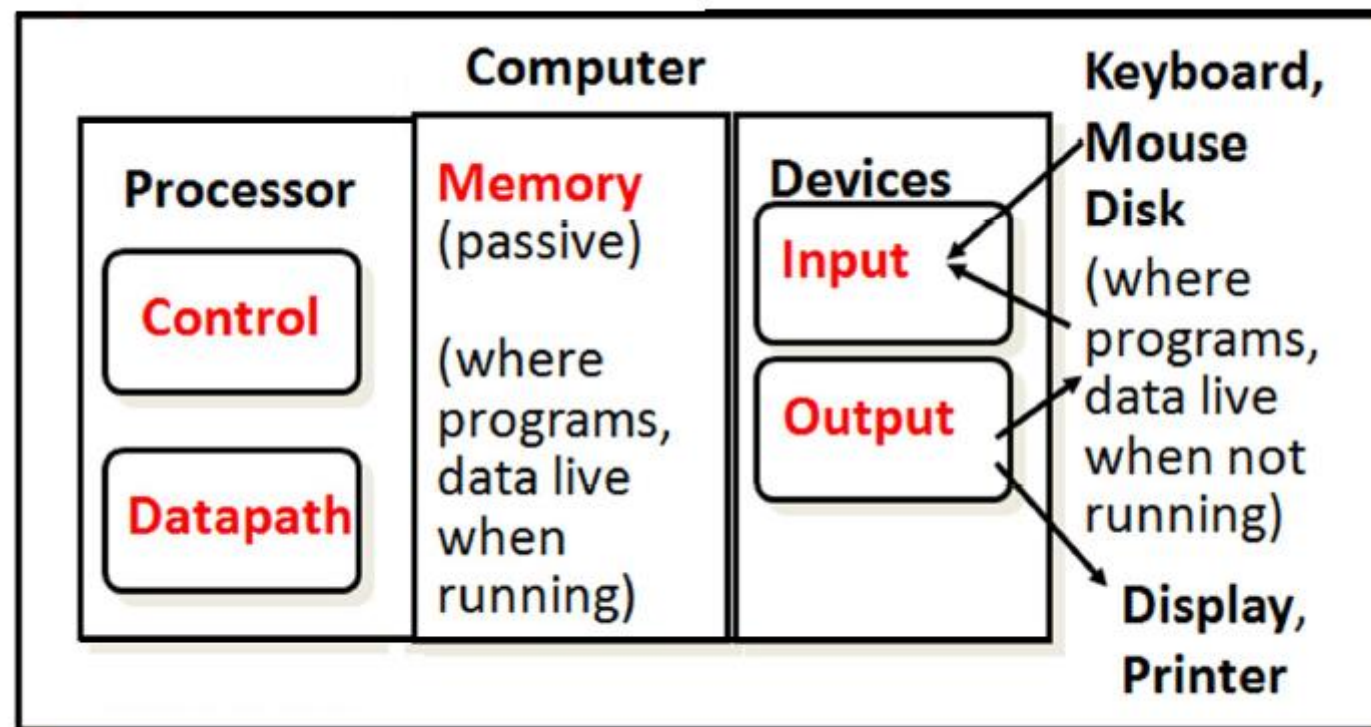
CH 6.1 - Instruction execution cycle and Stages of processor

- Instruction Execution Cycle
- Performance of Computer Systems
- Performance of CPU

Instruction Execution Cycle

Five Major Components of Computer Organisation:

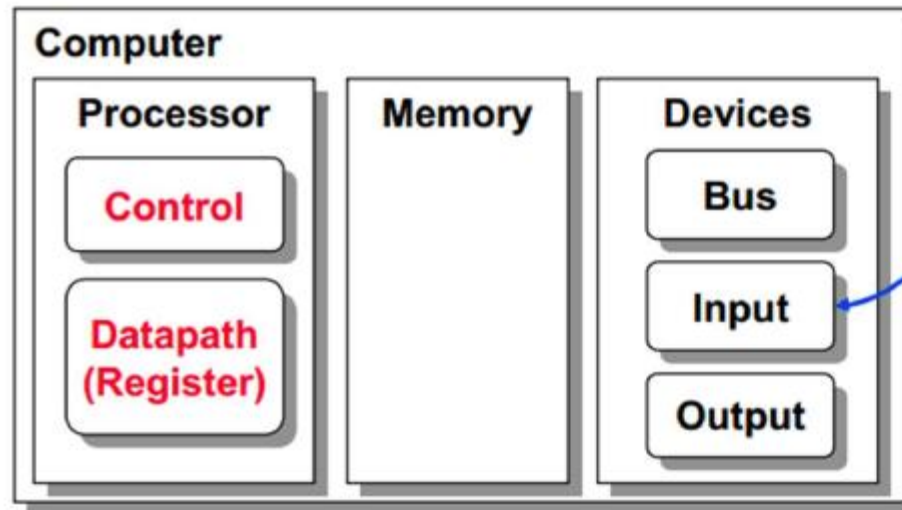
- Control unit
- Datapath
- Memory
- Input Devices
- Output Devices



Instruction Execution Cycle

Loading a Program via Input Device

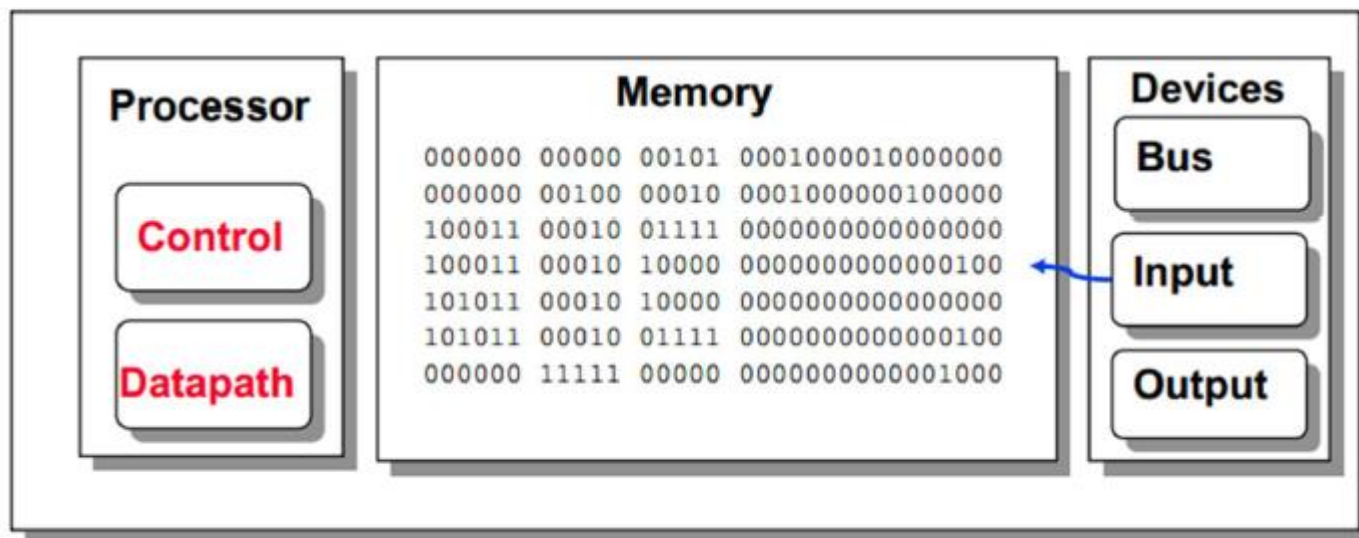
```
000000 00000 00101 0001000010000000
000000 00100 00010 0001000000100000
100011 00010 01111 0000000000000000
100011 00010 10000 00000000000000100
101011 00010 10000 00000000000000000
101011 00010 01111 00000000000000100
000000 11111 00000 00000000000001000
```



Instruction Execution Cycle

Storing a Program in Memory

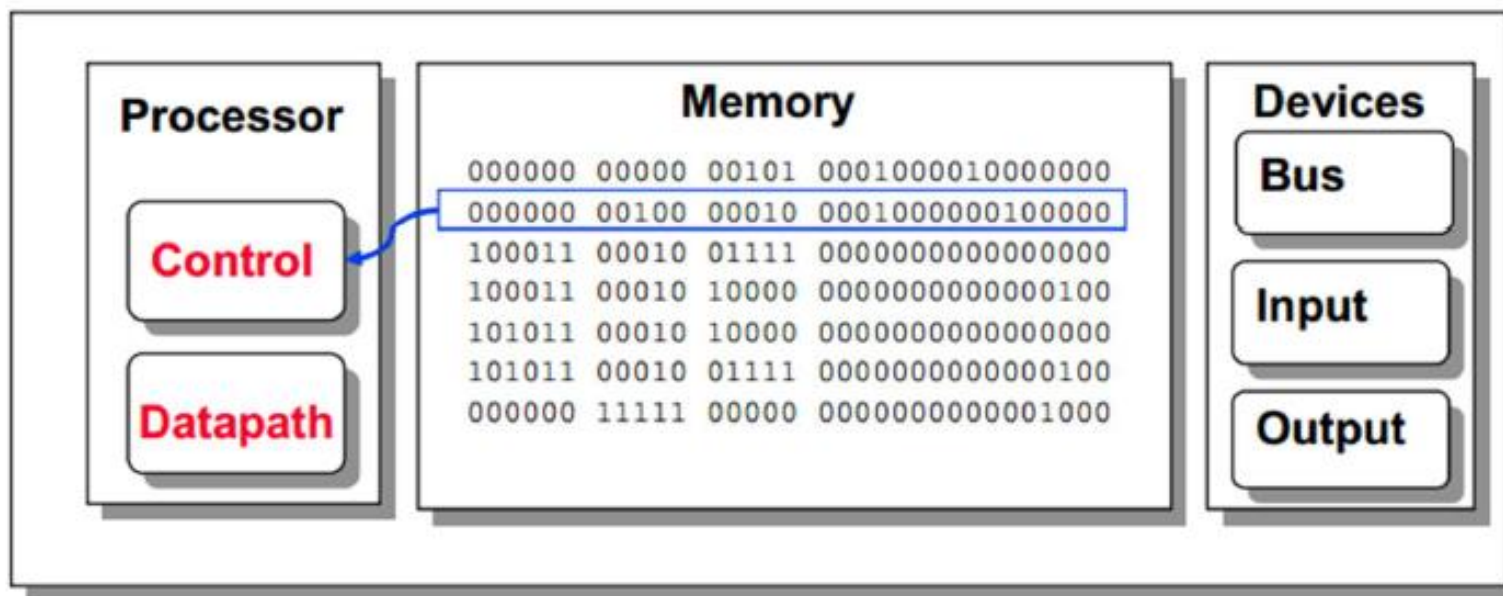
Memory stores **both** instructions and data



Instruction Execution Cycle

Execution Cycle - Instruction Fetch

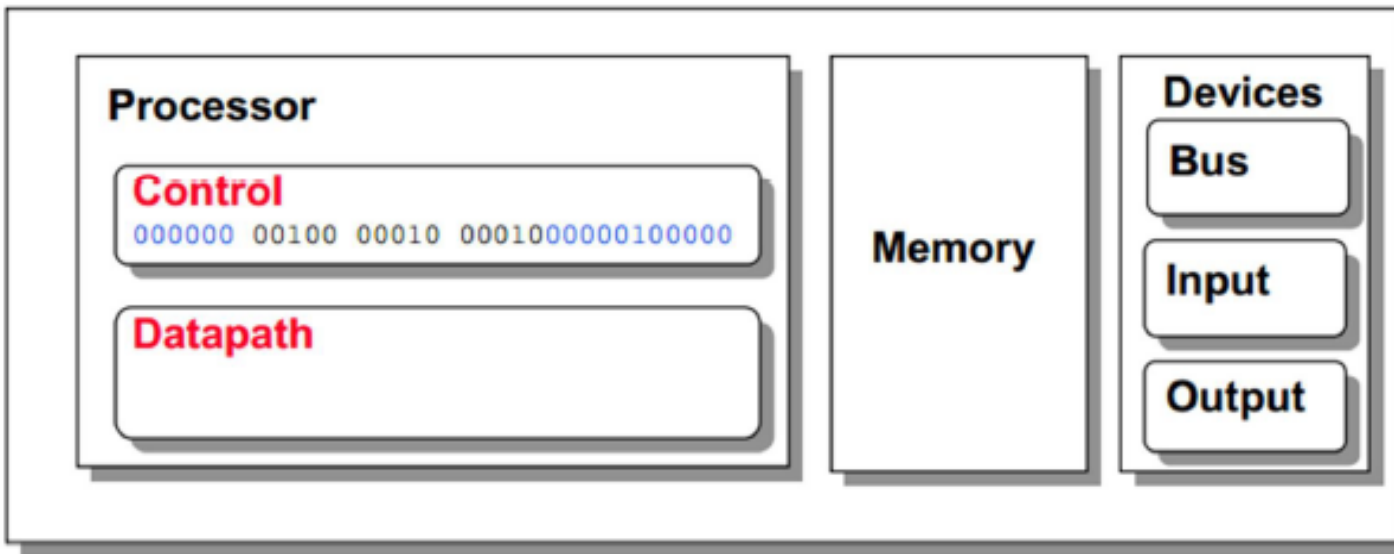
Processor **fetches** an instruction from memory



Instruction Execution Cycle

Execution Cycle - Instruction Decode

Control unit **decodes** the instruction to generate control signals that determine what to execute (determine the instruction format, instruction, and registers/operand being used).



Instruction Execution Cycle

Execution Cycle – Operand Fetch from Memory and Datapath Executes the Instruction

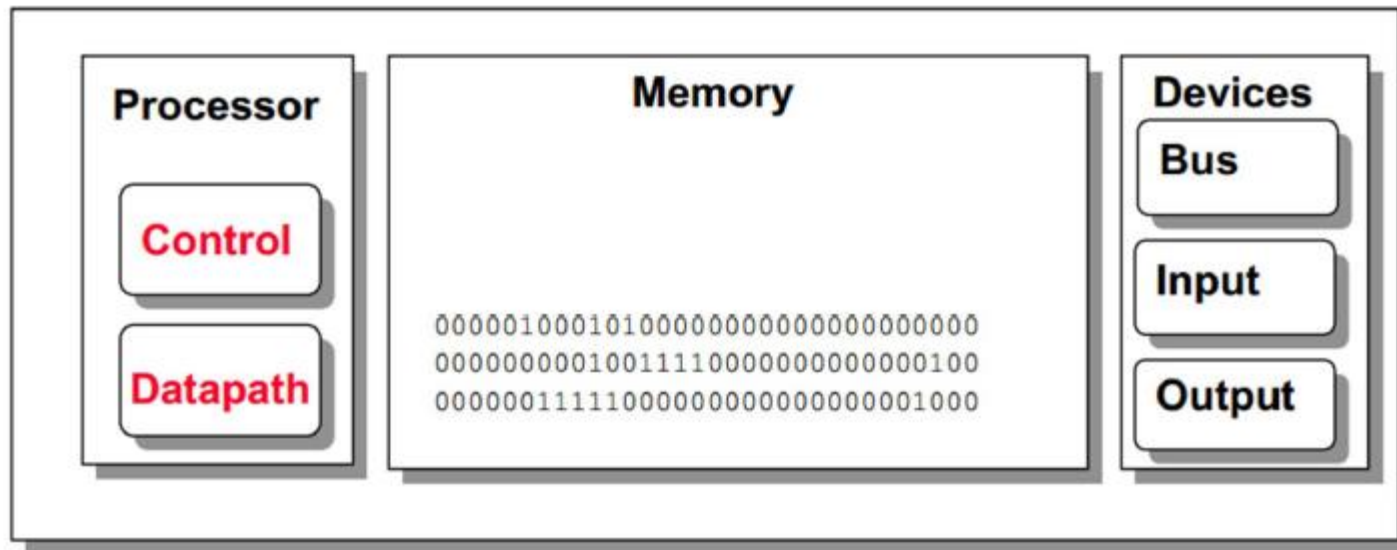
Datapath **executes** the instruction as directed by control



Instruction Execution Cycle

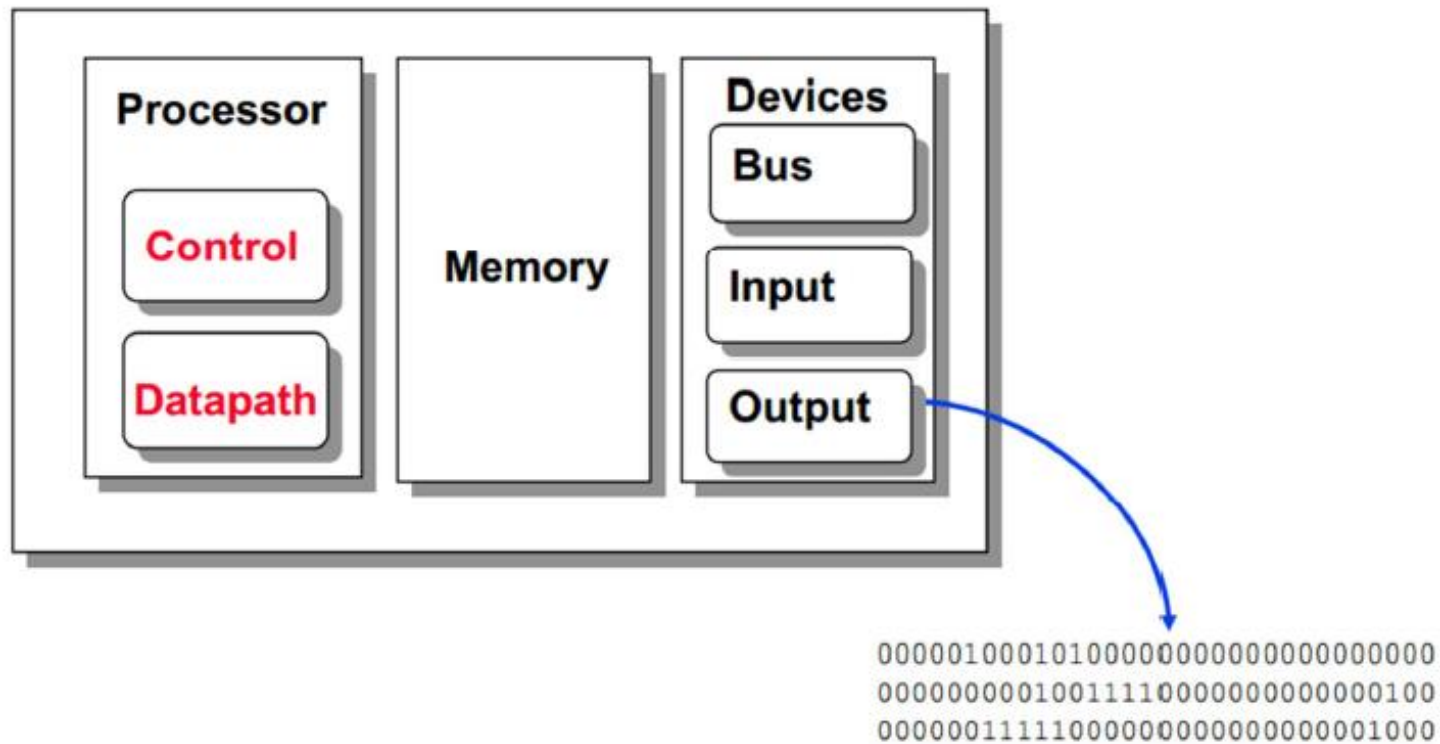
Execution Cycle – Result Store in Memory

At program completion, the data to be output resides in memory



Instruction Execution Cycle

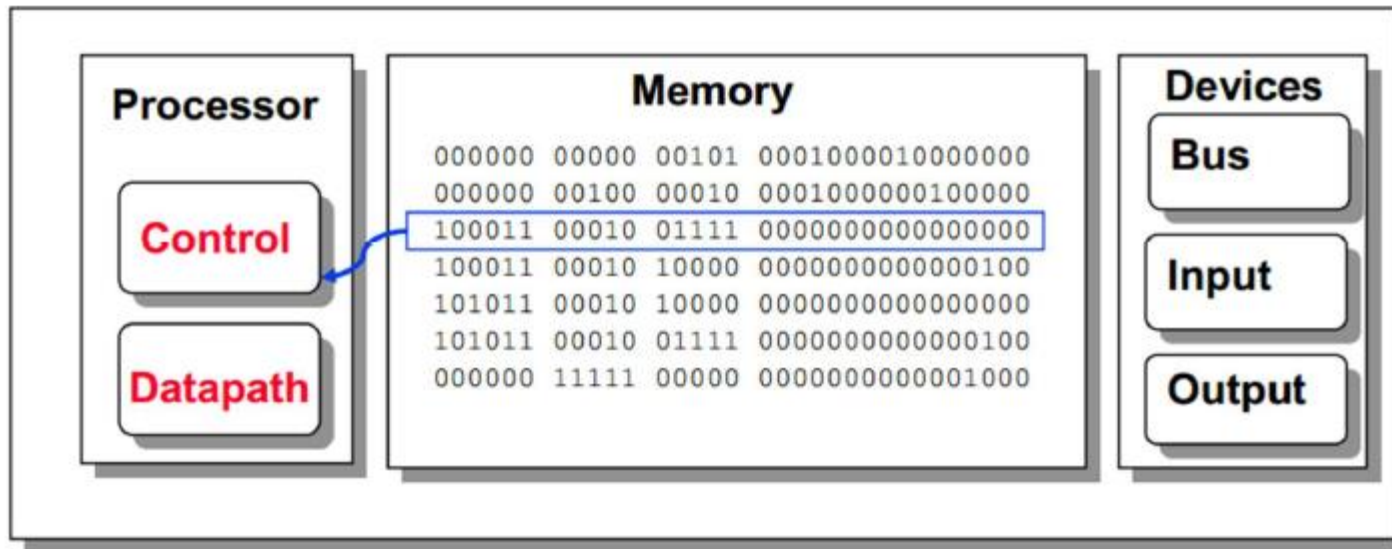
Data Output via Output Device



Instruction Execution Cycle

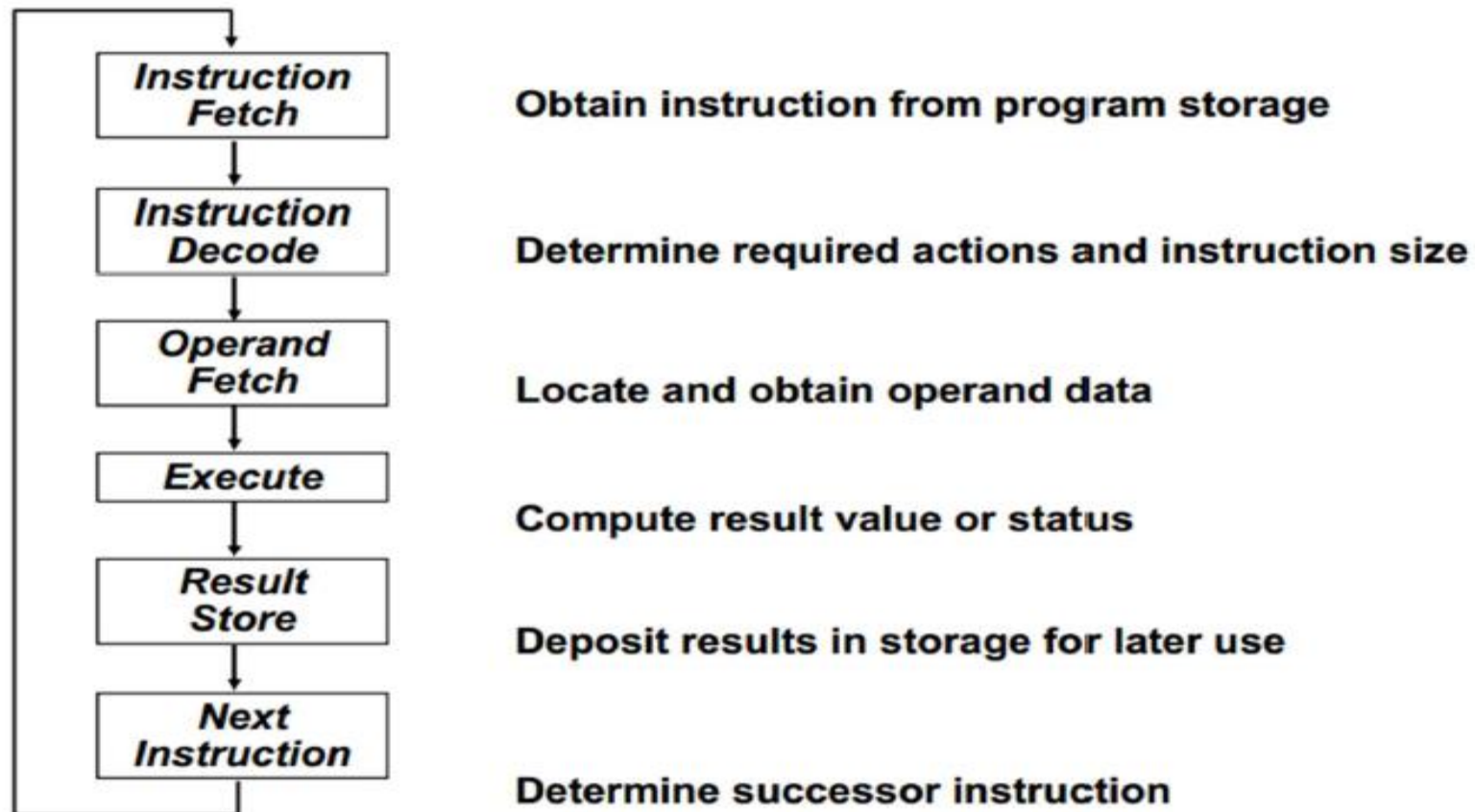
Execution Cycle Begins Again – Next Instruction Fetch

Processor **fetches** the next instruction from memory



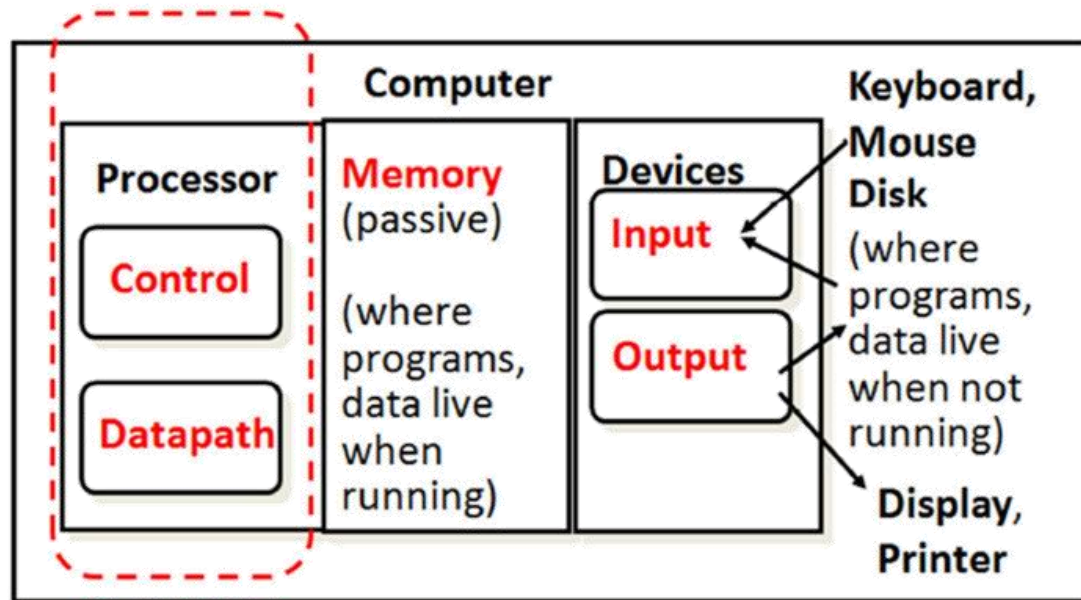
Instruction Execution Cycle

Machine Interpretation: Instruction Execution Cycle in Computer Organisation



Instruction Execution Cycle

In a processor, the **control unit** and the **datapath** are two essential components that work together to execute instructions

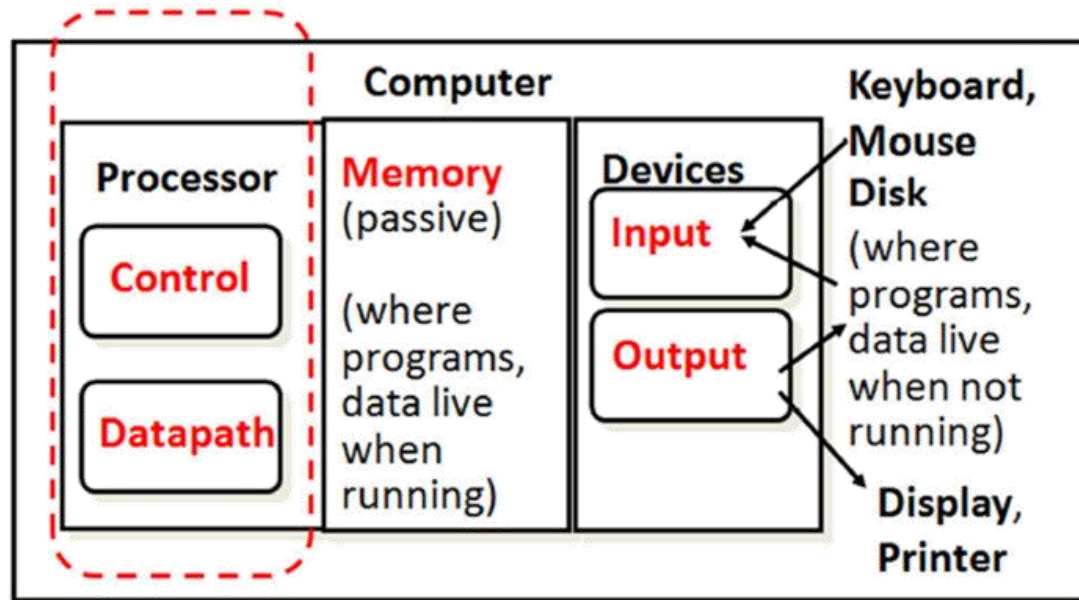


Control needs to have circuitry to:

- Decode the instruction
- Issue signals that control the way information flows between datapath components
- Control what operations the datapath's functional units perform
- Decide which is the next instruction and obtain it from memory

Instruction Execution Cycle

In a processor, the **control unit** and the **datapath** are two essential components that work together to execute instructions



Datapath needs to have circuitry to:

- Execute instructions – functional units (e.g., adder) and storage locations (e.g., register file)
- Interconnect the functional units accordingly
- Load data from and store data to memory

Performance of Computer System

- Basic performance measure:
 - **Response time/ Execution time:**
The time between the start and the completion of a task
 - **Throughput:**
The total amount of tasks done in a given time period
- Main factors influencing the performance
 - Processor and memory
 - Input/output controllers and peripherals
 - Compilers
 - Operating system
- Challenge is to satisfy constraints of:
 - Cost – Optimizing the design and components while keeping the overall cost within budget
 - Power – Minimizing power consumption while ensuring optimal performance
 - Performance – Maximizing system performance within the given cost and power constraints

Performance of CPU

Execution time or CPU time is used to measure CPU performance

Execution Time / CPU Time

= Instruction count $\times$ CPI $\times$ Clock cycle time

= (instructions/program) (cycles/instruction) (seconds/clock cycle)

CPI (Cycles Per Instruction)

- Average number of clock cycles per instruction for a program

Clock cycle time

- The time for one clock period
- Clock cycle = $1 / \text{clock rate}$

How to improve CPU time:

Instruction count: ISA and compiler technology

CPI: organisation, ISA and compiler technology

Clock rate: hardware technology and organisation

Performance of CPU

Example:

Consider the following performance measurements for a program:

Measurement	Computer A	Computer B
Instruction count	10 billion	8 billion
Clock rate	4 GHz	4 GHz
CPI	1.0	1.1

Which computer is faster?

Execution Time / CPU Time

= Instruction count $\times$ CPI $\times$ Clock cycle time

$$\begin{aligned}
 \text{Clock cycle time} &= 1/\text{clock rate} \\
 &= 1/4 \text{ GHz} \\
 &= 2.5 \times 10^{-10} \text{ sec}
 \end{aligned}$$

Solution:

$$\begin{aligned}
 &\text{Computer A execution time} \\
 &= (10 \times 10^9) \times 1.0 \times (2.5 \times 10^{-10} \text{ sec}) \\
 &= 2.5 \text{ sec}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Computer B execution time} \\
 &= (8 \times 10^9) \times 1.1 \times (2.5 \times 10^{-10} \text{ sec}) \\
 &= 2.2 \text{ sec}
 \end{aligned}$$

Computer B is faster



Q & A
Thank you